PHYSICS

Paper 5054/02
Paper 2 Theory

General comments

The overall impression of the examiners was that candidates were generally well prepared for the examination. Good candidates were readily able to score high marks and even weak candidates were able to show some understanding of the syllabus in every question. It is pleasing that the majority of the candidates produced clear and legible answers. The detail provided by candidates in answers to ***Section B*** was usually very good and it was encouraging to find that the majority of candidates started a calculation by writing down the equation that was relevant. Candidates should be encouraged not to start their answer by repeating the question.

There were no lined pages on the question paper this year as the questions filled the question booklet.

There was no evidence of candidates having too much or too little time. There were only occasional instances of misinterpretation of the instructions on the question paper, when some candidates answered all three of the ***Section B*** questions.

Comments on specific questions

Section A

Question 1

(a) Although most candidates realised that a force and a distance must be known to calculate power, few answers specified that the distance was the height of the stairs and that the force was the weight of the candidate. Most candidates correctly suggested that the time should be measured.

(b) There were many possible precautions, for example ensuring that the time is measured by several people and averaged, that the height is measured vertically or that the balance being used to measure mass has no zero error. Most answers were vague and merely suggested that the experiment should be “repeated” or “done accurately”.

(c) The formulae for power and work were well known and both marks were often scored. However a fairly common error was the suggestion that force is calculated by measuring the acceleration of the candidate as he runs up the stairs and the formula F=ma then used. There was no penalty if extra kinetic energy was included as the candidate accelerates up the stairs, as long as the major increase in potential energy had been included in the calculation.

Question 2

(a) The front wheel of the cyclist should be marked anywhere between 36 m and 50 m, but not at either of these positions. Many candidates had not realised that the cyclist is slowing down at 12 s and thus could not have travelled 50 m. Such candidates failed to earn one mark in **(a)** but could earn all the other marks in the question, even if their line in **(b)** did not show any deceleration at the end of the graph.

(b) Points were generally plotted well on the graph axes at 4, 8 and 12 s. However many candidates merely joined the origin to the first point using a straight line. The question states that the cyclist accelerates from rest and so the initial line should be curved. After 12 s the gradient of the line should decrease to show that the cyclist slows down. A correctly shaped curve from 12 to 16 s was drawn by the best candidates but was not required for full marks.

(c) The question required candidates to find an average speed and this is most simply calculated with the standard formula for speed using the distance travelled by the cyclist at 12 s. Some candidates measured the slope of the graph, found several speeds in an attempt to find an average or used 16 s instead of 12 s. They were able to earn one mark if their calculation showed clear evidence of the formula being used for speed.

Answer: **(c)** 3.0 m/s

Question 3

(a) Most candidates realised that as the coil turns the sides cut the lines of magnetic flux. A number of candidates confused the generator with a motor and described how current caused the coil to turn.

(b) The purpose of the slip rings is to allow the induced current to be connected to the load without the wires becoming tangled. Many candidates merely stated that the split rings cause the current to change direction every half turn or were used to support the coil.

(c) The majority of candidates did not produce an adequate definition of Lenz's law as applied to the direction of the induced current. A common error was to state the right hand rule for finding the direction of the induced current. This was not accepted as Lenz's law.

(d) Increasing the number of turns in the coil, increasing the strength of the magnet or using a larger coil are all methods of increasing the induced e.m.f.. However some candidates incorrectly suggested that increasing the current applied would increase the induced e.m.f.

Question 4

(a) It was pleasing to find many answers that correctly defined the critical angle as the angle of incidence in the denser or slower medium when the angle of refraction is 90°. Other definitions were possible and accepted in terms of, for example, the smallest angle of incidence for which the ray emerges from the denser medium. A few candidates described total internal reflection rather than the critical angle.

(b) The refractive index may be found from the critical angle or by using 63° as the angle of incidence and 40° as the angle of refraction. Errors were caused when candidates used the wrong angles entirely or if 40° was used as the angle of incidence. In either case candidates were required to state the formula that they used in the calculation and the correct formula, even with the wrong values, earned credit. The inverse of the accepted answer earned two of the three marks available.

(c) The vast majority of the candidates drew a correctly reflected ray.

Answers: **(a)** 46° **(c)** 1.39

Question 5

(a) The question implied that the two spheres are rubbed separately to obtain positive and negative charges on each. Many candidates suggested the two spheres are rubbed together and this was accepted as long as the positive sphere clearly lost electrons and the negative sphere gained electrons. A common error was to confuse the simple situation with charging by induction. Many candidates incorrectly suggested that positive electrons were involved.

(b) The general quality of the field patterns drawn was encouraging, with, in general, lines drawn from the positive to the negative charges. Some candidates drew all field lines as straight even through there should be some curvature of the lines at the top and bottom of the diagram.

(c) The formula relating charge, current and time was well known. However many candidates made arithmetical errors in the calculation involving negative powers of ten.

Answer: **(c)** 9.0×10^{-11}A

Question 6

(a) As many candidates found this section difficult, the meaning of live was allowed as the wire of high voltage, where energy or voltage arrived at the circuit or the dangerous wire. The neutral wire was zero or low voltage or the wire connected to earth at the generator or the wire that completes the circuit with the live wire to provide passage of current. Many candidates failed to distinguish between current and voltage, e.g. it was common to find answers such as "there is no current in the neutral wire or voltage flows in at the live wire". The meaning of earth was usually known, with many clear statements that the earth wire was connected to ground or the wire that allowed fault currents to pass safely to earth.

(b) (i) The answer to this section simply required candidates to realise that, as each of the lamps are switched on one by one, the current increases, the wire becomes hotter or electrons move faster. Many candidates did not explain what happened clearly enough or stated that the fuse would blow or the circuit melt, ignoring the simpler changes that would happen initially.

(ii) If a fuse of too high a rating is used in a plug the dangerous situation occurs when the current becomes too large and the fuse does not blow, enabling the cable to overheat, causing a fire or damaging an appliance connected to the cable. Many candidates merely suggested that the fuse would blow, stating that a melting fuse could cause a fire.

(c) The majority of candidates gave a good answer to this simple question in terms of either the conduction of current by water or by the electrocution that may be caused if hands are wet when handling an electrical plug.

Question 7

(a) There was a lack of familiarity with the use of the cathode-ray oscilloscope and there were fewer candidates than expected who showed two correct connections between the resistor and the Y input of the CRO. A common mistake was to connect the battery across the Y-input or to show connections to the Y-gain control. However some candidates showed great familiarity with the CRO and even ensured that the negative of the cell in their diagram was connected to earth, a point that was not required by the mark scheme.

(b) Many candidates were able to calculate the p.d. in **(b)(i)**, but then sometimes did not draw a line in **b(ii)** 1.5 squares above or below the original trace.

Answers: **(b)(i)** 6.0V

Question 8

(a) The question asked candidates to explain the meaning of radioactive decay and the answer required mention of the spontaneous or random decay of a nucleus or the emission of alpha, beta or gamma particles from the nucleus. There was often confusion between decay and fission and some candidates gave a definition of half life. Once the nucleus, rather than an atom, had been established as the basis of radioactivity then marks were scored but scoring two marks for this section was surprisingly rare.

(b) Many candidates correctly identified the background radiation as being responsible for the count rate before treatment. There were also those who suggested that the food itself contained radioactive material – which was accepted, as was the idea that the count before treatment is needed as a control to determine whether there has been any increase in the count caused by the treatment. In **(ii)** many candidates made the correct deduction that the food has not become radioactive because the average count is the same or that the difference in count is very small, caused by random fluctuations. A few candidates came to the correct conclusion without actually considering the data. For example, "The food does not become radioactive because gamma-rays are very penetrating and pass straight through the food" was a typical approach of such answers.

Section B

Question 9

(a) Full marks were rarely awarded for this section, although all candidates obtained some marks for the common sense approach that suggests evaporation increases with an increase in temperature or when a fan blows across the surface, whereas it decreases if a sheet of plastic covers some of the surface or if the tank is painted white. The explanation of these effects caused some problem. Some candidates took the plastic cover as covering the tank rather than the surface of the water but still obtained the explanation mark for suggesting that the evaporated water might condense and fall back into the tank. Poorer answers suggested that the plastic over keeps the water hotter or cooler. The most effective answer explaining the effect of the fan was to suggest that the humidity of the air above the water decreases allowing more molecules to escape or that the fan blows away some of the lower energy molecules that would return without the effect of a wind. Although most candidates recognised that the tank would receive less energy when painted white it was rare to find a candidate who suggested that heat radiation, light or infra red would be reflected. More usually candidates made vague suggestions that "heat" was not absorbed by the white surface. Candidates should be encouraged to distinguish clearly between conduction, convection and radiation where thermal energy is transmitted.

(b) The calculations in this section were generally done well and the formulae for density, specific heat capacity and power were well known. The weakest section tended to be **(i)** where candidates had to multiply the area of the tank by the decrease in height of the water. Many candidates incorrectly attempted to use the density of water at this stage. If a candidate used a wrong value obtained in **(i)** for subsequent calculations in the rest of this section full marks could be obtained in these later sections if there were no more errors.

Answers: **(b)(i)** $0.015\ m^3$ **(ii)** 15 kg **(iii)** 3.3×10^7 J **(iv)** 825 W

Question 10

(a) In **(i)** most candidates realised that pressure = force/area but did not define 1 Pa as the pressure exerted when a force of 1 N acts on an area of $1\ m^2$. The calculation in **(ii)** was well answered with most candidates using the correct weight of 600 N. The answer to **(iii)** was very simple but some candidates became confused and suggested that the pressure decreases.

(b) (i) Many candidates incorrectly assumed direct proportion between volume and pressure. Those who correctly stated the Boyle's law formula were generally successful but many candidates failed to give any assumption that they made in their calculation, e.g. that the temperature is constant or that no gas escapes from the tyre.

(ii) The kinetic theory explanation of pressure and its increase with temperature was well known. Some candidates failed to explain that pressure is caused by the impact of molecules on the sides of the container, in this case the tyre. Full answers mentioned not only the increase in speed or kinetic energy of the molecules but also that the molecules hit the walls more frequently or harder. Weaker answers merely suggested that the molecules hit more as the temperature is raised.

Answers: **(a)(ii)** 2.5×10^5 Pa **(b)(i)** $0.014\ m^3$

Question 11

(a) A sizeable minority of candidates incorrectly stated that the component C was a resistor, or even a capacitor rather than a fuse. Stating that the fuse controls the current (rather than limits it) was insufficient to gain full credit for the purpose of a fuse. Most candidates correctly identified component D as a switch, and gave a good description of the purpose, referring to switching the current or circuit on and off.

(b) The calculations in this section were well done by almost all candidates, although a common mistake was to merely divide the 0.42 A into two and state that this was the current in both lamps, suggesting that resistors in parallel have the same current. This is only true if the lamps are identical and the question clearly states that the lamps are different.

(c) (i) Almost all candidates drew the circuit diagram correctly, but in **(ii)** most candidates incorrectly suggested that the current in the series circuit is larger. It was not appreciated that the total resistance when two resistors are connected in series is greater than when they are connected in parallel.

(d) Many candidates knew that resistance is inversely proportional to area and deduced the doubling of resistance, but some obtained the correct result following unnecessary lengthy calculation. The concept of resisitivity is not included in the syllabus.

Answers: **(b)(i)** 0.25 A **(ii)** 0.17 A **(iii)** 960 Ω **(d)** 1920 Ω